NETWORK SCANNING & RECONNAISSANCE REPORT

Prepared by:

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Course:

B.E. Computer Science and Engineering(Cyber Security)

Tool Used:

Nmap 7.98

Target:

Own Local Network

Date:

03 October 2026

1.INTRODUCTION

Network reconnaissance is the process of collecting information about devices and services present on a network.

In this project, Nmap was used to perform reconnaissance on my own local network. The purpose was to identify active hosts, check TCP and UDP ports, detect services, and attempt operating system detection.

2.TOOLS USED

Nmap 7.98

Ubuntu/Linux

Terminal

3.NETWORK INFORMATION

Local IP Address: 192.168.113.118

Network:192.168.112.0/20

Gateway:192.168.112.1

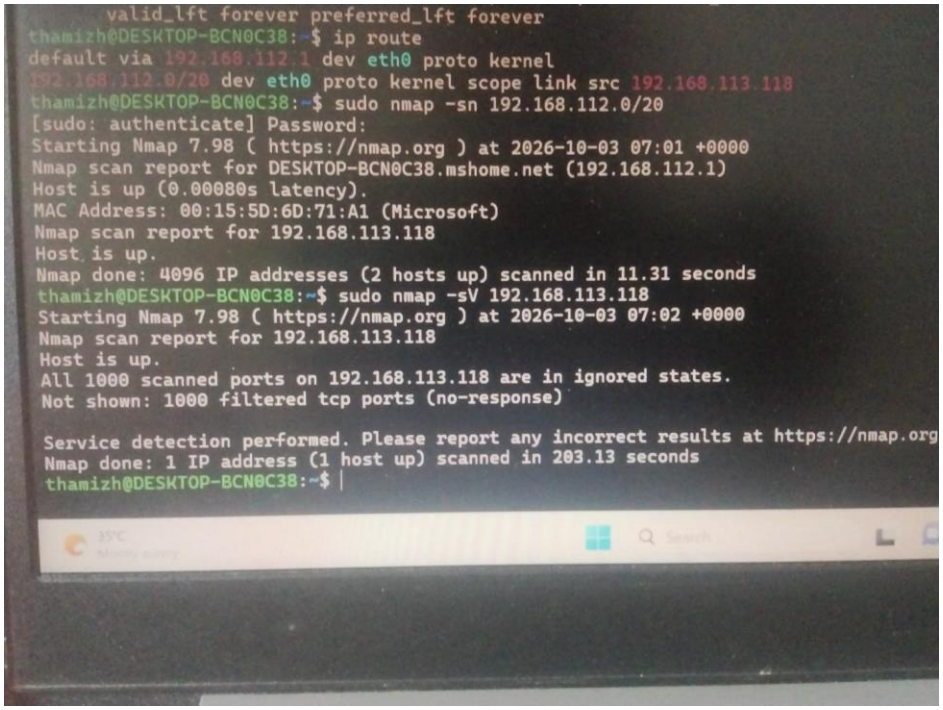
4.HOST DISCOVERY

Command used:

Sudo nmap –sn 192.168.112.0/20

Result:

The scan identified 2 active hosts in the local network. The target host 192.168.113.118 was found to be up.



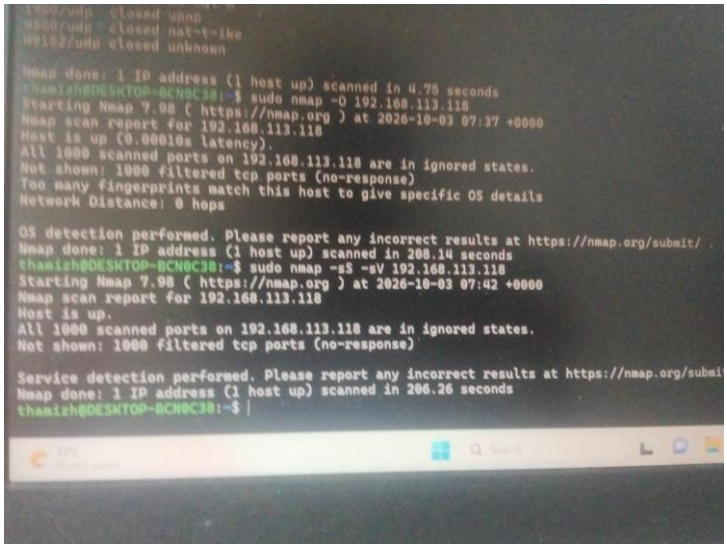
5.TCP PORT AND SERVICE SCAN

Command used:

Sudo nmap –sS –sV 192.168.113.118

Result:

The host was found to be up. All 1000 commanly scanned TCP ports were reported as filtered. No open TCP ports were identified in this scan.



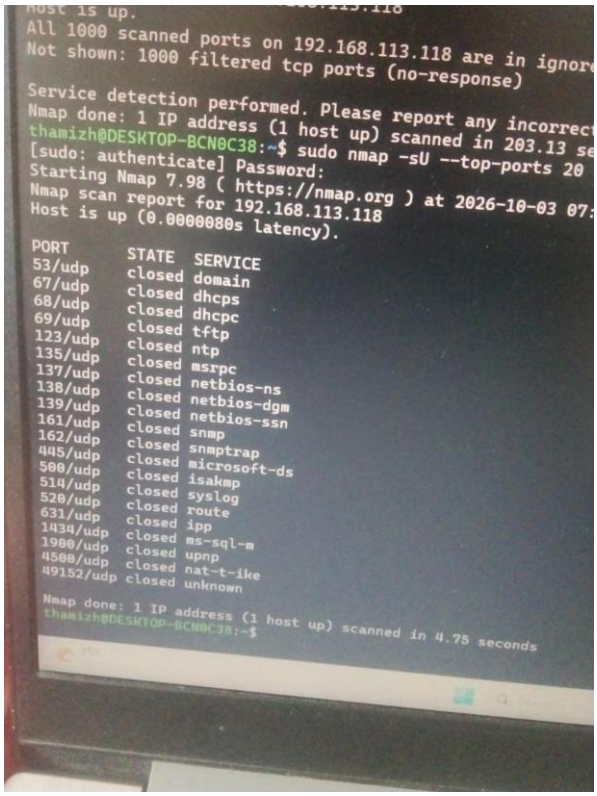
6.UDP PORT SCAN

Command used:

Sudo nmap –sU –top-ports 20 192.168.113.118

Result:

The scanned UDP ports were reported as closed. No open UDP services were identified in the top 20 UDP ports scanned.



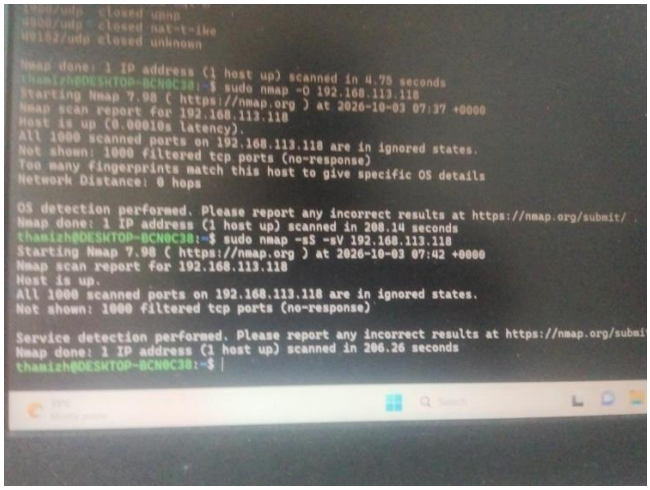
7.OPERATING SYSTEM DETECTION

Command used:

Sudo nmap –O 192.168.113.118

Result:

Nmap was unable to determine the specific operating system reliably because multiple OS fingerprints matched the host. Therefore, the operating system was not identified.



8.FINDINGS AND RESULTS

2 active hosts were identified during host discovery.

No open TCP ports were identified in the scanned ports.

The scanned top 20 UDP ports were closed.

TCP ports were reported as filtered, indicating that responses were blocked or not received.

Nmap could not reliably identified the operating system.

9.SECURITY RISKS

No open TCP or UDP services were identified in the scans performed. Therefore, no specific open services vulnerability was identified. However, filtered ports indicate that a firewall or network filtering mechanism may be blocking responses. If unnecessary services are enabled in the future, they should be disabled or properly secured.

10.SECURITY HARDENING RECOMMENDATIONS

Keep the operating system and software updated.

Enable and properly configure the firewall.

Disable unnecessary network services and ports.

Use storng passwords for user accounts.

Regularly monitor network activity and scan for unexpected open ports.

11.CONCLUSION

This network scanning exercise demonstrated how Nmap can be used to identify active hosts, scan TCP and UDP ports, detect services, and attempt operating system detection. The scan did not identify any open TCP or UDP services on the target host. The results also showed that the TCP ports were filtered. Regular scanning and proper security controls can help maintain a secure network.